MILITARY SPECIFICATION

CONTROL UNIT, COMMUNICATION SYSTEM C-10414(V)( )/ARC

This specification is approved for use by the Army Aviation Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1.0 SCOPE

1.1 Scope. This specification covers one type of control which provides control of inter-communication, audio levels, switching of airborne radio sets and navigation devices and is designated Control Unit, Communication System C-10414(V)( )/ARC. (See 6.1 and 6.4.)

2.0 APPLICABLE DOCUMENTS. The following document, of the issue as outlined in paragraph 2.1 shall form a part of this specification to the extent specified herein.

2.1 Issues of documents. The effective issue or revision of the following documents shall be that which is in effect on the date of invitation for bids or request for proposal, unless a specific issue is set forth in the cited specifications or contractual documentation.

SPECIFICATIONS

FEDERAL

NN-P-71	Pallets, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial).
PPP-T-0097	Tape, Packaging/Industrial, Filament Reinforced.
PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to U.S. Army Avionics Research & Development Command, ATTN: SAVAA-M, Fort Monmouth, New Jersey 07703 by using the self-addressed Standardization Document Improvement Proposal (DD-Form 1426) appearing at the end of this document or by letter.

AMSC N/A

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FSC 5821

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FEDERAL (Cont)

PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Boxes, Shipping, Fiberboard.
PPP-B-640	Boxes, Fiberboard, Corrugated, Triplewall.
PPP-S-760	Strapping, Nonmetallic (and Connectors).
QQ-S-781	Strapping, Steel, and Seals.

MILITARY

MIL-P-116	Preservation Methods of.
MIL-E-5400	Electronic Equipment, Aerospace, General Specification for.
MIL-F-7179	Finishes, Coatings, and Sealants for the Protection of Aerospace Weapons Systems.
MIL-P-7788	Panels, Information, Integrally Illuminated.
MIL-P-11268	Parts, Materials and Processes Used in Electronic Equipment.
MIL-M-13231	Marking of Electronic Items.
MIL-S-19500	Semiconductor Devices, General Specification for.
MIL-P-19644	Plastic Molding Material (Polystyrene Foam, Expanded Bead).
MIL-I-23053	Insulation Sleeving, Electrical, Heat Shrinkable, General Specification for.
MIL-C-25050	Colors, Aeronautical Lights and Lighting Equipment, General Requirements for.
MIL-L-25467	Lighting, Integral, Red, Aircraft Instrument, General Specification for.
MIL-P-26514	Polyurethane Foam, Rigid or Flexible, for Packing.
MIL-M-26542	Microphone, Dynamic, General Specification for.
MIL-P-55110	Printed Wiring Boards, General Specification for.

FEDERAL

FED-STD-356	Commercial Packaging of Supplies and Equipment.
FED-STD-595	Colors.

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MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-147	Palletized Unit Loads.

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MIL-STD-252	Classification of Visual and Mechanical Defects for Equipments, Electronic, Wire, and Other Devices.
MIL-STD-454	Standard General Requirements for Electronic Equipment.
MIL-STD-461	Electromagnetic, Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-471	Maintainability Verification/Demonstration/Evaluation.
MIL-STD-704	Aircraft, Electric Power Characteristics.
MIL-STD-781	Reliability Design Qualification and Production Acceptance Tests: Tests Exponential Distribution.
MIL-STD-785	Reliability Program for Systems and Equipment Development and Production.
MIL-STD-810	Environmental Test Methods and Engineering Guidelines.
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MS25212 (ASG)	Control Panel, Console Type, Aircraft Equipment, Basic Dimensions.

DRAWINGS

Avionics Research and Development Activity (57045)

DLD 6001858	Control Unit, Communication System, C-10414(V)1/ARC
DLD 6001859	Control Unit, Communication System, C-10414(V)2/ARC
DLD 6001860	Control Unit, Communication System, C-10414(V)3/ARC
DLD 6001873	Panel Illuminated RED, 28V, MX-10516/ARC
DLD 6001874	Panel Illuminated RED, 5V, MX-10515/ARC
DLD 6001876	Panel Illuminated GRN, 115V, MX-10517/ARC
DLD 6001872	Control Unit, Communication System, C-10414(P)/ARC

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3.0 REQUIREMENTS

3.1 Description. The Control Unit, Communication System C-10414(V)( )/ARC is a lightweight, selfcontained, solid state unit containing switching/mixing/amplifying circuitry designed to process audio output to an individual headset from up to five communications receivers and six navigation receivers. Each of the seven monitor switches has an integral audio level adjustment. A main volume control provides simultaneous audio level adjustment of the eleven selectable receiver circuits plus two direct inputs. Two additional direct audio inputs are provided without audio level control. A seven position switch selects up to five radio sets for transmission and reception, a private intercom line, and an interphone (ICS) position for

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use when two or more controls are interconnected within an airframe. A three-position toggle switch permits operating the ICS line in HOT-MIC or disabling the ICS audio input completely. A two-position flush-mounted, thumb actuated switch is used to select either a 5-ohm or a 150-ohm microphone. Simultaneously with selecting a 150 ohm microphone the headset output level will be reduced. The unit is flush-mounted by means of dzus fasteners located on the sides of the front panel mounting plate; operating controls and panel lamps are mounted on the front panel. The unit is capable of operating from a power source of 27.5-volts dc nominal. Panel lighting is operable from either +5 volts dc, 5-volts ac 400 Hz, 115-volt ac 400 Hz, or 27.5 volt-dc nominal depending upon panel selection. All field level adjustments and serviceable items shall be accessible without removal of cover plates and/or panels.

3.2 Construction. The units on order shall be constructed in accordance with the following drawing and data list:

DLD 6001858	Control Unit, Communication System, C-10414(V)1/ARC
DLD 6001859	Control Unit, Communication System, C-10414(V)2/ARC
DLD 6001860	Control Unit, Communication System, C-10414(V)3/ARC
DLD 6001873	Panel Illuminated RED, 28V, MX-10516/ARC
DLD 6001874	Panel Illuminated RED, 5V, MX-10515/ARC
DLD 6001876	Panel Illuminated GRN, 115V, MX-10517/ARC
DLD 6001872	Control Unit, Communication System, C-10414(P)/ARC

3.2.1 Weight (see 4.14.1). The weight of the completely assembled unit shall not exceed 2.4 pounds.

3.2.2 Outline dimensions (see 4.14). The dimensions of the completely assembled unit shall conform to the requirements of Military Standard MS 25212 (ASG) within the following limitations:

Panel/Housing Height - 2.625 inches (6.67 cm)

NOTE: Housing height behind front panel such that dimension .048 min. between top of panel and top of housing and between bottom of panel and bottom of housing.

Panel Width - 5.75 inches (14.60 cm)

Housing Width - 5.0 inches max. (12.77 cm)

Housing Depth - 4.625 inches (11.75 cm) (excluding front panel and rear connectors)

3.3 First article. When specified in the contract or purchase order, the contractor shall furnish First Article units as required (see 4.3 and 6.2d).

3.4 Finish (see 4.4). The case of the control shall be black, Color Number 37038 of FED-STD-595. The finishes and protective coatings used on all parts shall be in accordance with MIL-E-5400 and MIL-F-7179. Protective coatings shall be field restorable. All parts shall be suitably treated or finished so as to provide protection from corrosion in accordance with 3.1.2 of MIL-F-7179.

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3.5 Securing of parts (see 4.14).

3.5.1 General. Brackets, lugs, inserts, bolts, and other mounting arrangements shall retain items securely when subjected to specified service conditions.

3.6 Marking (see 4.4).

3.6.1 General. Marking shall conform to MIL-M-13231. Front panel marking shall be in accordance with equipment drawings.

3.6.2 Visibility. Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.6.3 Serial numbers. The following unit(s) shall be serial numbered.
Control Unit, Communication System, C-10414(V)1/ARC
Control Unit, Communication System, C-10414(V)2/ARC
Control Unit, Communication System, C-10414(V)3/ARC
Control Unit, Communication System, C-10414(P)/ARC
Panel Illuminated RED, 28V, MX-10516/ARC
Panel Illuminated RED, 5V, MX-10515/ARC
Panel Illuminated GRN, 115V, MX-10517/ARC

3.6.4 Front panels. (See 4.9.1)

3.6.4.1 Front panel (green). The unit(s) shall have an integrally illuminated front information panel. The panel shall have white legends against a black background for daylight operation and shall have green primary lighting for night operation, to provide compatibility with night vision goggles. Panel fabrication and performance shall be in accordance with Class 1-R, (except that the color is green) Type VI of MIL-P-7788. The panel lettering font shall be Alternate Gothic No. 3.

3.6.4.1.1 Chromaticity and Spectrum. The chromaticity shall fall within the area noted as aviation green of Figure 1, MIL-C-25050, (CEI chromaticity diagram), and with a dominant wavelength of 510 ± 15 nanometers and allowing no more than one percent of the relative peak amplitude to exist beteen 600 and 1000 nanometers. If filtering is necessary to meet requirements, measurements to assure acceptability shall be made subsequent to filtering.

3.6.4.1.2 Brightness. The maximum brightness shall be 1.0 ± 0.5 foot-lamberts.

3.6.4.2 Power. Red panel lighting shall be operable from the 28-volt DC or 5 volt DC/AC (400 Hz) aircraft source. Green panel lighting equipment shall be operable from the 115-volt AC (400 Hz) aircraft source. If incandescent light sources are utilized, by a change of lamp type, the unit shall be compatible with either a nominal 28-volt DC or a nominal 5-volt DC/AC (400 Hz) console panel lighting bus.

3.6.4.3 Panel protection. The CSC shall be designed so that each panel type, i.e., 5-volt, 28-volt, or 115-volt, receives its power over a separate power input pin, both at the input connector J1 and, at the panel connector. This is required to prevent damage to the CSC should it inadvertently be installed in the incorrect aircraft.

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The following pins are designated for panel illumination in input connector J1:

Panel Illumination Voltage	Pin Number
5 volts	R
28 volts	E
115 volts	J
Common	U

3.6.5 Alternate front panel red. (See 4.9.2) This paragraph shall be implemented only if contractually required. The units shall have an integrally illuminated front information panel. The panel shall have white legends against a black background for daylight operation and shall have red lighting for night operation. Panel fabrication and performance shall be in accordance with Class 1-R, Type V of MIL-P-7788. All lighting shall be aviation-instrument panel lighting red (@.620) as specified in MIL-C-25050 and shall meet the requirements of MIL-L-25467, MIL-P-7788 and MIL-STD-1472 and shall be externally controllable. By a change of panel, the unit(s) shall be compatible with either a nominal 28-volt DC or a nominal 5-volt DC/AC (400 Hz) console panel lighting bus. The front panel lighting shall be externally controllable so that the panel will be readable in direct sunlight (1×10^4 ft candles or in ambients as low as 10^{-5} foot candles). The intensity must be further downward controllable by external means so as to be legible and compatible with night vision aided devices.

3.6.6 Switch guard. The HOT MIC-NORM-ICS OFF Switch shall be provided with a guard.

3.6.7 Monitor switches. The integral monitor switch/level control shall be a combination push-pull (on-off) switch and rotary potentiometers. The switch shall be configured such that depressing the switch will activate that monitor line and pulling out on the switch will inactivate the monitor line. Minimum volume will be achieved at full counter-clockwise rotation of the potentiometer and maximum volume at full clockwise rotation.

3.6.8 Interface wiring. The assignment of interface connector pins shall be as specified in Table I. All audio inputs and outputs of the unit are designed for balanced operation with respect to signal ground, using twisted, shielded pair wiring. The unit shall be designed so that a single transmit-receive set unbalance occurring in the system wiring (audio lows shorted to ground) will not deleteriously impact on the operational and isolation integrity of the unit.

3.7 Parts, materials, and processes; general (see 4.14). In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirements therein, shall apply.

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3.8 Printed wiring assemblies (see 4.4). Printed wiring assemblies shall be processed for compliance with the requirements specified on the equipment drawings and MIL-P-55110 in order not to affect life and serviceability of the assemblies.

3.9 Semiconductors (transistors and diodes) (see 4.4). These electron devices (semiconductors and diodes) shall conform to MIL-S-19500 and the equipment drawings.

3.10 Tropicalization of materials (see 6.2.g). Materials shall be treated as follows:

3.10.1 Treating materials. Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material will be held to the practical minimum.

3.10.2 Toxicity. Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation, or maintenance of the equipment, or during use of the finished items when used for the purpose intended.

3.10.3 Flexibility. Treatment shall not affect the flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.11 Environmental service conditions. The equipment shall meet the operating and nonoperating service condition requirements of MIL-E-5400 for Class 1A equipment, except as specified below. The equipment shall be operable without degradation in specified performance before, during and after environmental testing unless otherwise specified herein. In the event of a relevent failure during an environmental test neccessitating a corrective action, the test will be repeated upon completion of the repair or modification.

3.11.1 Temperature-altitude (see 4.8.1). The equipment shall comply with the temperature-altitude requirements set forth in MIL-E-5400 for class 1A equipment. In addition, it shall withstand air transportation in unpressurized aircraft up to 50,000 feet above sea level. The input supply voltages shall be from 17.0 to 33.0-volts dc.

3.11.2 Humidity (see 4.8.2). The equipment shall be operable without degradation in specified performance, and shall sustain no physical damage, during and after prolonged exposure to extreme high humidity levels as encountered in tropical areas.

3.11.3 Dust (see 4.8.3). The equipment shall withstand, in both operating and nonoperating conditions, exposure to dust particles.

3.11.4 Salt fog (see 4.8.4). The equipment shall be resistant to the corrosive effects of salt-sea atmosphere.

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Table I. Pin Connections

J1 MS3112E20-41P	
Connector Pin	Function
C	28 VDC Power
D	Power Ground
R	Panel Light +5V
U	Panel Light Power Ground
e	Chassis Ground
H	Shields
L	Microphone Input (HI)
M	Microphone Input (LO)
K	Chassis Ground
b	Interphone Key
c	Key Common
d	Radio Key
f	Transmitter Control #1
g	Transmitter Control #2
h	Transmitter Control #3
r	Transmitter Control #4
t	Transmitter Control #5
X	Logic Select (PVT)
Y	Logic Select (ICS)
Z	Logic Select (XMIT 1)
i	Logic Select (XMIT 2)
j	Logic Select (XMIT 3)
k	Logic Select (XMIT 4)
m	Logic Select (XMIT 5)
n	Transmit Audio Out 1A
N	Transmit Audio Out 1B
P	Transmit Audio Out 2A
S	Transmit Audio Out 2B
T	Transmit Audio Out 3A
V	Transmit Audio Out 3B
W	Transmit Audio Out 4A
k	Transmit Audio Out 4B
s	Transmit Audio Out 5A
p	Transmit Audio Out 5B
q	PVT Hot Mic
A	Chassis Ground
B	Panel Light +28V
E	Spare
F	Spare
G	Panel Light 115V AC
J	Chassis Ground
a	

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Table I. Pin Connections (Cont)

J2 MS3112E20-41PW	
Connector Pin	Function
C	Shields
D	Headset Audio (HI)
E	Headset Audio (LO)
g	Private Line PVT-A
P	Private Line PVT-B
R	Interphone Line ICS-A
S	Interphone Line ICS-B
A	Receiver Audio In 1A
B	Receiver Audio In 1B
V	Receiver Audio In 2A
W	Receiver Audio In 2B
U	Receiver Audio In 3A
j	Receiver Audio In 3B
X	Receiver Audio In 4A
Y	Receiver Audio In 4B
k	Receiver Audio In 5A
m	Receiver Audio In 5B
F	Navigation Receiver Audio 1A
G	Navigation Receiver Audio 1B
a	Navigation Receiver Audio 2A
b	Navigation Receiver Audio 2B
n	Navigation Receiver Audio 3A
p	Navigation Receiver Audio 3B
c	Navigation Receiver Audio 4A
H	Navigation Receiver Audio 4B
J	Navigation Receiver Audio 5A
K	Navigation Receiver Audio 5B
s	Navigation Receiver Audio 6A
t	Navigation Receiver Audio 6B
N	Direct Audio (Uncontrolled) 1A
f	Direct Audio (Uncontrolled) 1B
q	Direct Audio (Uncontrolled) 2A
r	Direct Audio (Uncontrolled) 2B
L	Direct Audio (Controlled) 3A
M	Direct Audio (Controlled) 3B
d	Direct Audio (Controlled) 4A
e	Direct Audio (Controlled) 4B
Z	Chassis Ground
T	Chassis Ground
h	Chassis Ground
i	Chassis Ground

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3.11.5 Fungus (see 4.8.5). The equipment, in the assembled and ready for delivery condition, shall provide no nutrients in material, coating or contaminant for or support fungal growth when tested in accordance with 4.8.5. Only inherently fungus resistant grades of materials per requirement 4, MIL-STD-454 shall be used. Prior to delivery, all fungus nutrient contaminants or residues shall be removed.

3.11.6 Explosive atmosphere (see 4.8.6). The equipment shall not cause ignition of an ambient-explosive gaseous mixture with air when operated in such an atmosphere.

3.11.7 Rain (see 4.8.7). The equipment shall be operable without degradation in specified performance, and shall sustain no physical damage during periods of heavy precipitation with intermittent wind.

3.11.8 Acceleration (see 4.8.8). The equipment shall be operable without degradation in specified performance, and sustain no physical damage, when exposed to steady-state acceleration levels of up to 6g for a duration of 1 minute.

3.11.9 Vibration and shock.

3.11.9.1 Vibration (see 4.8.9.1). The equipment (with vibration isolators in place, if any) shall operate satisfactorily when subjected to vibration within the frequency range and amplitude as shown in Figure 2, Curve IIIb, Sheet 3 of MIL-E-5400, truncated at 500 Hz.

3.11.9.2 Shock.

3.11.9.2.1 Basic design (see 4.8.9.2.1). The equipment shall not suffer damage or subsequently fail to provide specified performance when subjected to 18 impact shocks of 15 g, consisting of three shocks in opposite directions along each of three mutually perpendicular axes. Each shock impulse shall have a time duration of 11 ±1 millisecond. The g-value shall be within ±10% when measured with a 0.2 to 250-Hz filter, and maximum g-value shall occur at approximately 5-1/2 milliseconds.

3.11.9.2.2 Bench handling (see 4.8.9.2.2). The equipment shall withstand shocks encountered in servicing.

3.11.9.2.3 Crash safety (see 4.8.9.2.3). The equipment shall withstand 12 impact shocks of 30 g's, consisting of two shocks in opposite directions along each of three mutually perpendicular axes. Each shock impulse shall have a time duration of 11 ± 1 milliseconds. The "g" value shall be within ±10 percent when measured with a 0.2 to 250 Hz filter and the maximum "g" value shall occur at approximately 5 ± 1/2 milliseconds. Bending and distortion shall be permitted, however, there shall be no failure to the attaching joints and the equipment or dummy load shall remain in place. Note: A dummy mass with outside structure and mounting points identical to the flight hardware and mass-loaded to be dynamically the same as the flight hardware may be used for the crash safety test.

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3.12 Voltage protection, low and reverse (see 4.8.11). The unit(s) shall be subjected to and meet the requirements of 3.13.1 and shall show no visual damage after being subjected to 33-volts dc reverse polarity and continuous operation at 17 to 33-volts dc.

3.13 Electrical requirements (see 4.10).

3.13.1 Input power (see 4.10.1). The unit(s) shall operate from normal 27.5-volt dc aircraft power as defined by MIL-STD-704, and shall maintain required performance over the input voltage range of 22.0 to 33.0 volts dc. It shall also maintain required performance when subjected to input overvoltage. During emergency mode, 17.0 volts to 22.0 volts, and undervoltage conditions, the unit will operate with degraded performance. When the input current is measured with 29.0-volt dc input, 1.25-volts audio output at the headset pins (before transmit attenuation (see 3.13.20)) and 0-dBm output on one transmitter audio output line, the CSC input current shall not exceed 250 milliamps, 7.25 watts. The input power requirements for front panel display lamps shall be no greater than:

a. 28-volts dc, 4.0 watts

b. 5-volts ac, 400 Hz and 5-volts dc, 4.0 watts

c. 115V ac, 400 Hz, 4 watts

3.13.2 Output interphone (see 4.10.2).

3.13.2.1 ICS line. With the selector switch set in any position, a 1.0-kHz signal at 0.6 volts applied to the required pad (see 4.7.f), which is connected to pins J1-L and J1-M (microphone input) and pin J1-b jumpered to ground, the output across pins J2-R and J2-S shall be between .35 and .43 volts; this output shall be the same at any position of the selector switch when the HOT MIC-NORM-ICS-Off switch is set to HOT-MIC and pin J1-b is ungrounded. HOT MIC and Interphone key (pin J1-b) operation shall be overridden when the selector switch is set in positions 1 through 5 and pin J1-d is jumpered to ground.

3.13.2.2 PVT line. With the selector switch set in PVT position, a 1.0-kHz signal at 0.6 volts applied to the required pad (see 4.7.f), to pins J1-L and J1-M (microphone input), the output across pins J2-g and J2-P shall be between .35 and .43 volts when pin J1-A is jumpered to ground (J1-C). Pin J1-A may be grounded in the interconnection wiring to provide automatic HOT MIC activation when selector switch is set to PVT.

3.13.2.3 ICS off. Any ICS signal to the headset shall be reduced at least 60 dB when switch is set to ICS OFF position.

3.13.3 Sidetone (see 4.10.3). With selector switch set to ICS position, HOT MIC-NORM-ICS OFF switch to NORM position, pin J1-d or J1-b jumpered to ground and 1.0-kHz signal at 0.6 volts applied to the required pad, the output voltage across pins J2-D and J2-E shall be -4.0 to -6.0 dB below the level obtained across J2-D and J2-E when a 0.39-volt rms, 1.0-kHz signal is applied to pins J2-R and J2-S. A comparable sidetone reduction shall occur when selector switch is set to PVT and pin J1-A is jumpered to ground.

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3.13.4 Response, microphone amp (see 4.10.4). The frequency response shall not vary more than +1.0 dB within the freq range of 700 Hz to 2250 Hz. The 3 dB ± 1 dB BW shall be 350 Hz to 4500 Hz. The 1 dB and 3 dB ± 1 dB levels are referenced to the level at 1kHz. The amplitude between 350 Hz and 4500 Hz shall be greater than 1 dB above the amplitude at 1 kHz. The slope of the asymptotic attenuation curve above 4500 Hz and below 350 Hz shall be no less than 6 dB per octave.

3.13.5 Distortion, microphone amp (see 4.10.5). The total harmonic distortion shall not exceed three (3) percent with a signal level of 0.6 volts applied to the required pad and five (5) percent with a signal level of 4.0 volts applied to the required pad over the frequency range of 350 to 4500 Hz. Center clipping ("zero-crossing distortion") shall be no greater than 2 dB with a signal level of 0.085 volts at 1.0 kHz applied to the required pad.

3.13.6 Attack and release time, AGC (see 4.10.6). With 1.0 kHz applied to pins J1-L and J1-M, the attack time shall be not more than 0.1 seconds and the release time shall be not more than 3.0 seconds ±1.5 seconds.

3.13.7 Breakpoint and slope, AGC (see 4.10.7). With 1.0 kHz applied to the pad connected to pins J1-L and J1-M (see 4.7.f) and when the linearity of the input-output curve varies 1.0 dB (this will be the breakpoint), the input signal level shall be between 0.2 and 0.5 volts with an increase in the input signal level from 0.5 to 5.0 volts, the slope of the input-output curve from the breakpoint shall be 2 dB maximum change in output for a 20 dB change in input. This requirement applies to the equipment operated in both the 5 ohm and 150 ohm microphone input positions.

3.13.8 Transmit audio output, microphone amp (see 4.10.8). With the microphone selector switch in position 1 and the channel selector switch in positions 1 through 5 and a 1.0 kHz signal at a level of 0.6 volts applied to the 5000-to-5-ohm pad connected to the microphone input, the output on the indicated output lines shall be 0.39 volts ±10%. With the microphone selector switch in position 2 and a 1.0-kHz signal at a level of 0.6 volts applied to a 2.4K to 150 ohm pad with blocking capacitor, 47 ufd or greater connected to the microphone input, the output on the indicated output lines shall be 0.39 volts ±10%.

3.13.9 Audio output, headset amp (see 4.10.9). The Headset amp shall contain an internally adjustable control to permit adjustment of the output level of the headset amp. With an input of 2.75V, 1 kHz at any one radio, NAV and DIRECT channel or an input of 0.39V, 1 kHz at either the ICS or PVT interphone input and the MIC switch in position 1 (5 ohms), the output across 8 ohms shall be adjustable from below 1.0 volts to above 1.6 volts. With the MIC switch in position 2 (150 ohms) after the output level was previously set to 1.25 volts with MIC switch in position 1, the output level shall be 1.25V ± 10%.

With the MIC switch in position 1 (5 ohms) and the control set for 1.25 volts output across 8 ohms the amplifier output voltage shall be limited when the input voltage to a radio NAV or DIRECT channel cross a threshold between 2.75 and 3.87 volts. The limiter circuitry shall operate effectively through 5.47 volts of input, to limit the output to within 4 dB of the maximum set value of 1.25V. The output level of a received tone shall not be reduced by more than 9 dB by the presence of another tone greater than the limiting level at any RADIO, NAV, or DIRECT channel.

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3.13.10 Response, headset amp (see 4.10.10). The frequency response shall not vary more than ±1.0 dB within the freq range of 700 Hz to 2250 Hz. The 3 dB ± 1 dB BW shall be 350 Hz to 4500 Hz. The 1 dB and 3 dB levels are referenced to the level at 1 kHz. The amplitude between 350 Hz and 4500 Hz shall not be greater than 1 dB above the amplitude at 1 kHz. The slope of the asymptotic attenuation curve above 4500 Hz and below 350 Hz shall be no less than 6 dB per octave.

3.13.11 Distortion, headset amp (see 4.10.11). When frequencies over the range of 350 to 4500 Hz are applied with the required input voltages to audio input lines, the total harmonic distortion at the headset amp output shall not exceed ten (10) percent. Center clipping ("zero-crossing distortion") shall be no greater than 2 dB below the peak output level with a signal level of 0.39 volts at 1.0 kHz applied to the audio input lines.

3.13.12 Broadband Noise, Headset (see 4.10.12). The headset broadband noise output of the unit(s) shall not exceed 0.9 millivolts when subjected to the test of 4.10.12.

3.13.13 Isolation, monitor switches/audio lines (see 4.10.13). With MIC switch in position 1 (5 ohms) and with 1.0 kHz applied to the specified audio inputs and a 1.25V output at the headset amplifier, the output shall not exceed 1.25 mv when the monitor switch is set to the "Off" position and the selector switch is set to any other audio input line. With 2.75 volts, 1.0 kHz applied to any audio input line, monitor switches to "ON" position, the audio level on any other audio line shall be less than 2.75 millivolts. (When the input signal is applied to one audio input, the signal shall be less than 27.5 millivolts on the two common NAV audio lines.)

3.13.14 Isolation, transmitter lines (see 4.10.14). With 2.75 volts, 1.0 kHz level applied to any monitor input line or 0.39 volts, 1.0 kHz level applied to the ICS or PVT lines and the headset amplifier output set for 1.25V, the undesired voltage level at each transmitter audio output or control line shall be at least 120 dB below the normal 0.39 volts output, except that on the selected transmitter audio output line the undesired signal shall not be detectable in the output noise floor as observed using a 5 Hz bandwidth. The output noise floor as observed using a 5 Hz bandwidth shall be no greater than -90 dB referenced to 0.39V out. Similarly, a 1.0 kHz, 0.39 volt, signal on any selected ICS, PVT or transmit audio line shall have an isolation of at least 120 dB below normal, 0.39 volts output at any unselected transmit audio line.

3.13.15 Transient protection (see 4.10.15). The unit(s) shall meet specified requirements after subjection to a +80-volt pulse and a -80-volt pulse for 0.2-second duration three (3) times within 1 minute. In addition, the unit shall meet specified requirements after subjection to a ±600-volt spike of 5-microseconds ±0.2 microseconds duration, applied three (3) times within 1 minute.

3.13.16 Headset amp, unterminated (see 4.10.16). The headset amp output shall be stable (no oscillation) when pins J2-D and J2-E are open circuited, or short circuited and grounded, and the monitor and selector switches are placed to any position and all volume controls are rotated to any level between minimum and maximum audio output.

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3.13.17 Transmitter control (see 4.10.17). With selector switch set to positions 1 through 5, pin J1-d jumpered to ground, and a dc voltage applied to specified pins, the transmitter control lines shall withstand 1.0 amps through the lines without any degradation.

3.13.18 Function selector logic (see 4.10.18). With selector switch to positions PVT, ICS, 1,2,3,4, and 5, a separate 1-ampere capacity circuit pin to ground shall be provided. The normally open pin shall be grounded (DC Power Grd) when its respective selector switch position is selected.

3.13.19 Microphone power (see 4.10.19). The unit shall provide up to 10 mA at 10-volts dc ±20% on the microphone high input (J1-L) with power return through the microphone low input (J1-M) when the microphone selection switch is set to position 2.

3.13.20 Received audio input attenuation during transmit (see 4.10.20). With a 0.39 volt rms, 1.0 kHz input applied to PVT or ICS input or a 2.75 volt rms, 1.0 kHz input applied to any other audio input and the headset amplifier output set for 1.25V rms, the received audio inputs appearing at the headset amplifier output shall be selectively attenuated 12 dB ± 1 dB (below the 1.25Vrms level) as shown in Table II when selector switch is set to positions 1 through 5 and pin J1-d is grounded.

3.13.21 Audio level (volume) adjustments (see 4.10.21).

3.13.21.1 Main volume control. With all volume controls set at maximum audio output and the input to a specific audio input line adjusted to obtain 1.25 volts output at the headset, the output shall be reduced by 40 ± 3 dB when the main volume control is set for minimum audio output.

3.13.21.2 Monitor audio level adjustment. With all volume controls set at maximum audio output and the input to a specific audio input line adjusted to obtain 1.25 volts output at the headset, the output shall be reduced by 27 ± 2dB when the individual audio level control (for the selected radio audio monitor input) is set for minimum audio output. The audio outputs of all other monitor lines shall be uneffected by this adjustment. This requirement shall be met for any headset amplifier adjustment level between 1.1 VRMS and 1.7 VRMS.

3.13.22 TEMPEST. The equipment shall comply with the requirements of Confidential Appendix A, titled "TEMPEST Requirements" for the Control Unit, Communication System C-10414(V)( )/ARC.

3.14 Electromagnetic interference (see 4.11). The communications system control units shall comply with the requirements of Table 1 of MIL-STD-461A Notice 4 as defined herein after:

CE01	CS01	RE02	RS02
CE02	CS02	RE02.1	RS03.1
CE03	CS06		
CE04			
CE05			

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Table II. Received audio input attenuation

SELECTOR SWITCH POSITION	RECEIVED AUDIO LEVEL								DIRECT RADIOS
	PVT	ICS	1	2	3	4	5	NAV 1	NAV 2	
1	-12 dB	-12 dB	0 dB	-12 dB					-12 dB	0 dB
2	-12 dB		-12 dB	0 dB	-12 dB				-12 dB	0 dB
3	-12 dB			-12 dB	0 dB	-12 dB			-12 dB	0 dB
4	-12 dB				-12 dB	0 dB	-12 dB		-12 dB	0 dB
5	-12 dB					-12 dB	0 dB	-12 dB	-12 dB	0 dB
PVT	Sidetone	0 dB								0 dB
ICS	0 dB	Sidetone	0 dB							0 dB

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3.14.1 Audio susceptibility, CS01 (see 4.11.1). The unit(s) shall comply with CS01 requirements. The unit(s) headset (J2-D and J2-E) output shall not exceed 4.0 MV between 30 Hz and 50 kHz. The Interphone (J2-R and J2-S) output shall not exceed 13 MV between 30 Hz and 50 kHz.

3.14.2 Conducted susceptibility, CS02 (see 4.11.2). The requirements of CS02 shall apply except that all RF signals be modulated 50 percent AM at 1 kHz. The success criteria for this requirement shall be a maximum noise output for the Headset Amplifier (J2-D and J2-E) as follows: 4.0 MV between 150 kHz and 199 MHz, and between 251 MHz and 400 MHz; 4.0 MV between 200 MHz and 250 MHz. The maximum noise output for the Interphone Terminals (J2-R and J2-S) shall be 13.0 MV.

3.14.3 Radiated susceptibility, RS03 (see 4.11.3). The requirement of RS03 shall apply except that all RF signals shall be modulated 50 percent AM at 1.0 kHz. The field strength for RS03 shall be as specified in MIL-STD-461 except that it shall be increased to 10 volts/meter over the range 2 to 30.0 MHz. The success criteria for this requirement shall be a maximum noise output of 4.0 MV for the Headset Amplifier (J2-D and J2-E) and 13.0 MV at the Interphone Terminals (J2-R and J2-S).

3.14.4 Radiated susceptibility (induction field (see 4.11.4). The unit(s) Headset noise output (J2-D and J2-E) shall not exceed 10 MV and the Interphone noise output (J2-R and J2-S) shall not exceed 20 MV when subjected to the test of 4.11.4.

3.14.5 Bonds, grounds, connectors and cables (see 4.14). Bonding for the equipment current path return, RF potentials and shock hazard shall be so installed that expansion, contraction or movement incident to normal service use will not break or loosen the connection. Surface preparation for bonds and grounds shall be accomplished by removing all anodic film, grease, paint, lacquer or other high resistance properties from the immediate area of contact. Bonding impedance shall not exceed 2.5 milliohms. Case and chassis mounted connectors shall have approved RF conductive finish and shall be installed as above. Bonding impedance of the connector shell to the case shall not exceed 2.5 milliohms. Interconnecting cables requiring shields shall have a minimum of 40-dB shield attenuation to electric fields over the frequency range of 10 kHz to 400 MHz. The shield shall be constructed of tinned copper braid. Connectors with cables having shields shall have an approved RF conductive finish and provided with EMI backshells for terminating the shields.

3.15 Burn-in (see 4.6). Burn-in shall be performed as described in paragraph 4.6.2.

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3.16 Reliability program. The unit(s) shall be subjected to a reliability program that shall provide for, but not be limited to, the following reliability tasks as described in MIL-STD-785:

a. Reliability Management

b. Parts Reliability

c. Reliability Demonstration

d. Failure Data

3.17 Reliability (see 4.6). The equipment minimum Mean-Time-Between Failures (MTBF, θ_1) shall be 3500 hours when used in the following manner:

Ambient Temperature: -54°C to +55°C

Duty Cycle: 100 percent (5 minutes receive, 1 minute transmit)

Environment: Airborne

This requirement shall be demonstrated in accordance with paragraph 4.6.3.

3.18 Operational inspection (see 4.13). Each unit shall be operated as specified after applicable processes and inspections are completed to determine that each unit is operable prior to packaging. Voice quality shall indicate proper operation of the unit(s).

3.19 Interchangeability (see 4.12). Like units, assemblies, subassemblies, and replaceable parts shall conform to Requirement 7 of MIL-STD-454 and shall be physically and functionally interchangeable without modification of such items or of the equipment.

3.20 Systems safety engineering (see 4.14).

3.20.1 Personnel hazards. Personnel hazards shall be kept to a minimum through compliance of configuration changes and parts selection with Requirement 1 of MIL-STD-454. Compliance with these requirements will be verified through a visual inspection.

3.20.2 Edge rounding exposed. Exposed edges shall be rounded to a minimum radius of 0.04 inch (1 mm), and exposed corners to a minimum of 0.5 inch (13 mm).

3.20.3 Radioactive materials. Radioactive materials shall not be used (e.g., luminous dials/markings, electron tubes, surge arrestors and lenses).

3.21 Workmanship (see 4.14). Workmanship shall be in accordance with Requirement 9 of MIL-STD-454.

4.0 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all

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inspection and test requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection and test requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections and tests set forth in the specification where such inspections and tests are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First Article inspection (does not include packaging (see 4.3)).

b. Inspection covered by subsidiary documents (see 4.4).

c. Quality conformance inspection.

(1) Quality conformance inspection of equipment before packaging (see 4.5).

(2) Quality conformance inspection of packaging (see 4.15).

4.3 First article. Unless otherwise specified in the contract or purchase order, the First Article inspection shall be performed by the contractor.

4.3.1 First article units. The contractor shall furnish that number of First Article units of the complete C-10414(V)( )/ARC as specified in the contract or purchase order.

4.3.2 First article inspection. The first article inspection shall consist of the inspections specified in subsidiary documents covering the items listed in 4.4, burn-in/reliability specified in 4.6, and the inspections specified for Group A, Group B, and Group C (see Tables III, IV, and V, respectively). The inspection shall be performed in the following order: 4.4, 4.6, Group A and Group B for all units, and designated units for specific subgroups within Group C in accordance with 4.5.4.1. After completion of Group C environmental tests, conforming units shall be reinspected and shall pass all Group A inspections.

4.4 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection before preparation for delivery.

Item	Where Required
Finish	3.4
Marking, Lighting, and Interface	3.6
Printing Wiring Assemblies	3.8
Semi-conductors	3.9

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4.5 Quality conformance inspection of equipment before packaging. Each unit of equipment shall be subjected to Burn-In (3.15) prior to inspection. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.5. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. Failure to meet any requirement as a result of Burn-In shall be classified as a defect and recorded as such. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government at its discretion, may perform all or any part of the specified inspections, to verify the contractor's compliance with specified requirements. Test equipment for Government verification inspection shall be made available by the contractor.

4.5.1 Group A inspection. Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in Table III. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined here, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subjected to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels, and AQL's indicated in Table III. Group A inspection shall be performed in any order which is satisfactory to the Government, except that the operational inspection (4.13) of the entire lot (including the sample units) shall be performed last. Subsequent handling of the units shall be minimized to reduce possibility of damage to the units.

4.5.2 Group B inspection. This inspection, including sampling, shall conform to Table IV and to the procedures for small sample inspection of MIL-STD-105 using the special inspection levels. The AQL shall be 6.5 percent defective and the inspection level shall be S4 for normal, tightened and reduced inspection. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection.

4.5.2.1 Subgroup I. All units selected for group B tests will be subjected to subgroup I tests.

4.5.2.2 Subgroup II. Four units of these selected for group B tests will be subjected to subgroup II tests.

4.5.3 Order of inspection within Group B. Group B inspection shall be performed in any order which is satisfactory to the Government.

4.5.4 Group C inspection. This inspection shall be as listed in Table V and shall be performed on lots that have been subjected to and met Group A and Group B inspections. No defects shall be allowed. Sample unit(s) shall be selected in accordance with 4.5.4.1.

4.5.4.1 Sampling for inspection. Units shall be selected at random for each group of inspections in Table V as specified in 4.5.4.1.2. The first samples selected shall be from the first quality conformance inspection production lot(s). Selection of these samples shall be as specified and within 30 calendar days (45 calendar days for the fungus test). After selection, the samples shall have been subjected to and completed the Group C inspections (see 6.3).

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Table III. Group A Inspection

Inspection	Req. Para.	Insp. Para.	AQL Major	AQL Minor
Visual and Mechanical	3.20 3.21	4.14		
Printed Circuits Assemblies:				
Transmit Circuit Card Assy			0.65%	2.5%
Receive Circuit Card Assy			0.65%	2.5%
Interconnect Board			0.65%	2.5%
C-10414(P)/ARC (less front panel)			1.0%	4.0%
C-10414(V)( )/ARC			1.0%	4.0%
Electrical:				
Output, interphone (to include ICS, PVT and ICS OFF)	3.13.2	4.10.2	*1.0% for the combined group	
Sidetone	3.13.3	4.10.3		
Response, microphone amp	3.13.4	4.10.4		
Distortion, microphone amp	3.13.5	4.10.5.1		
Transmit audio output, microphone amp	3.13.8	4.10.8		
Audio output, headset amp	3.13.9	4.10.9.1		
Response, headset amp	3.13.10	4.10.10		
Distortion, headset amp	3.13.11	4.10.11.1		
Transmitter control	3.13.17	4.10.17.2		
Function Selector Logic	3.13.18	4.10.18.2		
Microphone Power	3.13.19	4.10.19		
Received Audio Input attenuation during transmit	3.13.20	4.10.20		
Audio Level (Volume) adjustments	3.13.21	4.10.21		
Operational inspection (with Test Set)	3.18	4.13.1	1.0%	

*All electrical defects are considered major

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Table IV. Group B Inspection

Inspection	Req. Para.	Insp. Para.	AQL
Subgroup I:			
Input power	3.13.1	4.10.1	6.5% for the combined group
Distortion, microphone amp	3.13.5	4.10.5.2	
Attack and release time, AGC	3.13.6	4.10.6	
Breakpoint and slope, AGC	3.13.7	4.10.7	
Audio output headset amp	3.13.9	4.10.9.2	
Distortion headset amp	3.13.11	4.10.11.2	
Broadband noise, headset	3.13.12	4.10.12	
Isolation, monitor switches/ audio lines	3.13.13	4.10.13	
Isolation, transmitter lines	3.13.14	4.10.14	
Transient protection	3.13.15	4.10.15	
Headset amp, unterminated	3.13.16	4.10.16	
Transmitter control	3.13.17	4.10.17.1	
Function selector logic	3.13.18	4.10.18.1	
Subgroup II:			
Interchangeability (Mechanical)	3.19	4.12	6.5%
*Interchangeability (Electrical)	3.19	4.12	6.5%
Operational Inspection (system simulation)	3.18	4.13.2	6.5%

*Inspection shall be performed prior to assembly into the units.

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Table V. Group C Inspection

	REQ. PARAGRAPH				INSPECTION PARAGRAPH
	UNIT 1	UNIT 2	UNIT 3	UNIT 4	
Subgroup I:					
Weight		3.2.1			4.14.1
Voltage protection		3.12			4.8.11
Temperature-altitude	3.11.1				4.8.1
Humidity	3.11.2				4.8.2
Vibration			3.11.9.1		4.8.9.1
Shock			3.11.9.2.1 & 3.11.9.2.2		4.8.9.2
Dust			3.11.3		4.8.3
Subgroup II:					
Electromagnetic interference		3.14			4.11
Explosive atmosphere		3.11.6			4.8.6
Acceleration			3.11.8		4.8.8
Fungus			3.11.5		4.8.5
Salt Fog		3.11.4			4.8.4
Rain	3.11.7				4.8.7
Voltage - Front Panel (Green)		3.6.4			4.9.1
Voltage - Front Panel (Red)		3.6.5			4.9.2
Crash Safety			3.11.9.2.3		4.8.9.2.3
Subgroup III:					
TEMPEST				3.13.22	4.10.22

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4.5.4.1.1 Subgroup I. For this subgroup, two (2) units from the first eight (8) production units of the first 500 or part thereof on order shall be selected for the first Group C inspection. If quantities in excess of 500 are on order, two (2) units from each successive 500 shall be selected for Group C inspection.

4.5.4.1.2 Subgroup II. For this subgroup, two (2) units from the first eight (8) production units shall be selected for the first Group C inspection. If quantities in excess of 1000 are on order, two (2) units from each successive 1000 shall be selected for Group C inspection. These samples may be the same units selected for Subgroup I tests.

4.5.4.1.3 Subgroup III. For this subgroup, one (1) unit from the first eight (8) production units of the first 1000 or part thereof on order shall be selected for the first Group C inspection. If quantities in excess of 1000 are on order, one (1) unit from each successive 1000 shall be selected for Group C inspection.

4.5.4.2 Order of inspection within Group C. Group C inspection shall be performed in an order which is satisfactory to the Government.

4.5.4.3 Group C failures. Actions required relative to Group C failures shall be as specified in the contract.

4.5.5 Reinspection of conforming Group C sample units. Unless otherwise specified, lots which have been subjected to and passed Group C inspection may be delivered on the contract or order, provided all damage is repaired, cleaned of all foreign matter, and the sample units are resubjected to and pass Group A inspection.

4.6 Burn-in and reliability.

4.6.1 Conditions precedent to burn-in and reliability testing. Prior to doing any burn-in or reliability acceptance testing, the contractor shall perform a thermal and vibration survey in accordance with the paragraphs titled "Thermal Survey" and "Vibration Survey" of MIL-STD-781.

4.6.2 Burn-in. Each deliverable equipment shall be subjected to a 48-hour on-time burn-in period, the last 24 hours of which shall be failure free. Each unit which fails during the last 24-hour period shall be repaired and returned to test until it successfully survives a 24-hour period without failure. Failures which occur during burn-in test shall be noted and reported, but will not be counted towards the establishment of the equipment MTBF. Duty cycle imposed, vibration, environmental test profile and parameters to be measured shall be those included in the contractor's approved reliability test procedure.

4.6.3 Reliability testing.

4.6.3.1 The system reliability shall be demonstrated in accordance with Test Plan XVIIC, MIL-STD-781 on a minimum sample of two equipments (see Table VI). For this test plan, the MTBF (θ_1) shall be 3500 hours.

4.6.3.2 Reliability qualification test conditions. The reliability qualification test conditions shall be in accordance with Appendix B, the paragraphs titled "Combined Environments for Turboprop Aircraft" and "Helicopter Equipment" of

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Table VI. Test sequence for limited sample size

UNITS AVAILABLE

3			2		1
Unit 1	Unit 2	Unit 3	Unit 1	Unit 2	Unit 1
Temperature-ALT Rain	Explosion	Dust	Temperature-ALT Rain	Dust Explosion	Temperature-ALT Dust Explosion Rain
Humidity	Salt fog	Acceleration Vibration Shock Fungus	Humidity Salt fog	Acceleration Vibration Shock Fungus	Humidity Fungus (2) Salt fog (2) Acceleration Vibration Shock

NOTES:

(1) The tests above the broken line are considered non-damaging. The order of these tests and choice of units to use may be varied if convenient.

(2) Following fungus and salt fog testing, the test item shall be thoroughly washed, cleaned, dried, and refurbished, if necessary, before proceeding with subsequent tests.

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MIL-STD-781. A failure is defined as the inability of the equipment to perform its required function. This includes catastrophic failures as well as any event which causes any departure from performance as required by this specification. Proper instrumentation shall be provided to insure immediate recognition of failure. Voltage cycling shall be according to the paragraph titled " Electrical Stress", Appendix B.

4.7 Standard test conditions. Unless otherwise specified herein, inspections shall be performed under test conditions of GENERAL REQUIREMENTS, Test Conditions, of MIL-STD-810 and the following conditions:

a. Input power: 27.5 ± 0.5-volts dc.
b. Operational duty cycle: One (1) minute talk, five (5) minute listen.
c. Warm-up time: Not more than 2 minutes.
d. Output loads: Headset amp; 8 ohms. Transmit audio output; 150 ohms (see 4.7g).
e. Input loads: Audio lines and interphone; 150 ohms (see 4.7g).
f. Input load microphone; MIC switch in position 1, 5,000 to 5-ohm pad; MIC switch in position 2, 2.4K to 150 ohm pad with blocking capacitor, 47 ufd or greater (see 4.7g).
g. Connector (J1) Pins: Pin pairs N-P, S-T, V-W, k-s, p-c are each terminated with a 150-ohm resistor. When MIC switch is in position 1, pins L and M should be terminated with a 5K to 5-ohm resistor pad with the junction of the 5K and 5-ohm resistors connected to pin L. When MIC switch is in position 2, Pins L and M should be terminated with 2.4K and 150 ohm pad with blocking capacitor, 47 ufd or greater, with the junction of the 2.4K and 150-ohm resistors connected to Pin L through the blocking capacitor. Pins b, c, and d, are connected to a single pole, double-throw switch, with pin c connected to the center pole. All shields are connected to pin H (chassis ground).
h. Connector (J2) Pins: Pin pairs A-B, V-W, U-j, X-Y, k-m, s-t, F-G, a-b, n-p, c-H, J-K, d-e, L-M, q-r, N-f, R-S, g-P are each terminated with a 150-ohm resistor. Pin pair D and E are terminated with an 8-ohm resistor. Shields should be tied to pin C.

4.8 Environmental test procedures. The unit(s) shall be subjected to the following test methods of MIL-STD-810, and other tests as specified to determine compliance with service condition requirements of 3.11. Electrical tests performed before, during and after environmental testing shall include those tests required to determine compliance with paragraphs 3.13.2, 3.13.4, 3.13.5, 3.13.8, 3.13.9, 3.13.10, 3.13.11, 3.13.16, 3.13.17, and 3.13.18.

4.8.1 Temperature-altitude. The unit(s) shall be subjected to the test of Method 504.1, Procedure 1, for equipment Category 4. In addition, the procedure shall include a 1-hour exposure (non-operation) to a pressure of 3.4 inch of Hg, simulating air-transport at 50,000 feet.

4.8.2 Humidity. The unit(s) shall be subjected to the test of Method 507.1, Procedure II for compliance with 3.11.2. In Step 6 of Procedure II, measurements shall be taken during the last five hours of the last cycle.

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4.8.3 Dust. The unit(s) shall be subjected to the test of Method 510.1, Procedure 1, with the exception that the air velocity in Steps 1, 2 and 3 shall be 200 + 100 FPM. Step 3 may proceed immediately after stabilization in Step 2 if desired.

4.8.4 Salt fog. The equipment shall be subjected to the test of Method 509.1 of MIL-STD-810. Electrical operation shall not be required. All external knobs and controls shall be operated after the test. There shall be no sticking or binding of any part, and there shall be no corrosion of finishes and metals. Such corrosion shall be defined as any visible degradation of the test item surfaces that can be attributed to flakey, pitted, blistered, or otherwise loosened finish or metal surface.

4.8.5 Fungus. The equipment model in the assembled and as ready for delivery condition, shall be subjected to 28 days of Method 508.1, MIL-STD-810, except that the cyclic conditions listed under the paragraph titled "Incubation of Test Items" shall not be used. Instead, the conditions of 29°C (84°F) and 85% RH should be maintained throughout the test. The equipment test model shall not be specially cleaned for the fungus test, except for the cleaning it receives during or after production. If it is necessary for the manufacturer after assembly to remove accumulated production and handling contaminants by cleaning prior to packaging and delivery, the equipment shall not be tested for three days after cleaning to allow for complete evaporation of the cleaning compound. All enclosed or gasketed assemblies shall be opened and the interior sprayed with the specified mixed spore suspension. After 14 and 28 days of test, the control items per the paragraph titled "Control Items", of Method 508.1, MIL-STD-810 shall show profuse growth over at least 50% of the area of the control items. After 28 days, based upon visual examination, the equipment assembly shall pass if it shows no more than only sparse microbial growth with restricted tuberal growth development in an area of 1 to 10% or less of the total areas and no more than six unrelated minute colonies with mycelial development in areas only in other than critical circuit portions; such as terminal spacings, printed circuit boards, etc., with sparse growth due to random contamination or traces of unmixed material ingredients. The equipment shall be operated during the inspections step (the paragraph titled "INSPECTION", Method 508.1, MIL-STD-810). The equipment shall fail if it does not meet operational performance criteria or if it shows more than the growth specified above.

4.8.6 Explosive atmosphere. The equipment shall be subjected to the test of Method 511.1, Procedure I of MIL-STD-810.

4.8.7 Rain. The equipment with cables in place shall be exposed to a "steady state" rainfall of 4 + 1 inches per hour on all surfaces of the test items for a total time duration of not less than 2 hours. After completion of the rain test, the units shall be removed from the test chamber and turned on within 10 minutes after test completion. The units shall be operated and tested to determine satisfactory system operation.

4.8.8 Acceleration. The equipment shall be subjected to the test of Method 513.2, Procedure II, of MIL-STD-810. The test level shall be 6g for at least 1 minute.

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4.8.9 Vibration and shock.

4.8.9.1 Vibration. The equipment when normally mounted (with vibration isolators in place, if any), shall be subjected to the test of Method 514.2, Procedure I, Category C, curve M, of Figure 514.2-3 (Part 1), with exception of the frequency range, which shall be limited to 5-500-5 Hz. The time schedules shall be in accordance with Table 514.2-III, except that the sweep time corresponding to Curve M shall be 30 minutes. The equipment shall be operated during the test. Also prior to, and following these tests, the equipment shall be operated in accordance with MIL-STD-810, General Requirements, "Performance of Test". The equipment shall not have any degradation in performance and shall not suffer any mechanical damage.

4.8.9.2 Shock.

4.8.9.2.1 Basic design. The equipment shall be subjected to the test of Method 516.2, Procedure I of MIL-STD-810. The shock pulses shall be in accordance with Figure 516.2-2 for flight vehicle equipment.

4.8.9.2.2 Bench handling. The equipment shall be subjected to the test of Method 516.2, Procedure V, of MIL-STD-810.

4.8.9.2.3 Crash safety. The equipment shall be subjected to the test of Method 516.2, Procedure III, figure 516.2.2 of MIL-STD-810, for half-sine 30g, 11-ms curve, and meet the requirements of paragraph 3.11.9.2.3.

4.8.10 Test sequence. To eliminate cumulative effects of potentially damage inducing tests on a single test item, separate samples should be used for each test whenever possible. The test sequence of Table VI shall be used for limited sample size.

4.8.11 Voltage protection low and reverse. Apply plus 17 and 33-volts dc to pins J1-C and J1-D for 5 minutes at each voltage, and then apply 33-volts dc with plus to pin J1-D and negative to pin J1-C for 1 minute. The unit(s) shall meet the requirement of 3.12.

4.9. Front Panels.

4.9.1 Front panel (green). With an adjustable 115 volt, 400 Hz ac power source applied to pins J1-J and J1-U, the unit(s) shall meet the requirements of 3.6.4.

4.9.2 Alternate front panel (red). With an adjustable 28 volts dc or an adjustable 5 volts dc/ac (400 Hz) power source applied to pins J1-E and J1-U or pins J1-R and J1-U, respectively, the unit(s) shall meet the requirements of 3.6.5.

4.10 Electrical test.

4.10.1 Input power. With the unit(s) operating under standard test conditions except 29.0 Vdc shall be applied to Pins J1-C and J1-D. Adjust the input signal level at pins J2-R and J2-S for 1.25 volts across J2-U and J2-E. Jumper Pin J1-d

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to ground and apply 0.6 volts at 1.0 KHz to a 5000-ohm to 5-ohm pad connected to Pins J1-L and J1-M. While these inputs are applied, measure the dc current to determine compliance with 3.13.1. Reduce the input voltage to 21 Vdc at pins J1-C and J1-D. Verify that the performance requirements of paragraphs 3.13.2, 3.13.5, 3.13.8, 3.13.9, 3.13.11, and 3.13.18 are met. Raise the input voltage to 33 Vdc at pins J1-C and J1-D. Verify that the performance requirements of paragraphs 3.13.2, 3.13.5, 3.13.8, 3.13.9, 3.13.11, and 3.13.18 are met. Connect 28 Vdc to pins J1-R and J1-U and measure the dc input current to determine compliance with 3.13.1.

4.10.2 Output, interphone.

4.10.2.1 ICS. Set MIC switch in position 1 (5 ohms). Apply 0.6 volts at 1.0 kHz to a 5,000 to 5 ohm pad connected to pin J1-L, M and set selector switch to ICS. Jumper pin J1-d to ground and measure the output voltage at pins J2-R,S. Switch jumper to pin J1-b and measure the output voltage at all positions of selector switch. Jumper pin J1-d to ground and J1-b to ground simultaneously and measure the output voltage with selector switch in positions 1 through 5. Remove both jumpers and set Hot MIC switch to HOT MIC position and measure output voltage with selector switch in position 1 through 5. With Hot MIC switch in Hot MIC simultaneously jumper J1-d to ground and measure output with selector switch in position 1 through 5. These measurements shall be made to determine compliance with 3.13.2.1.

4.10.2.2 PVT line. Apply 0.6 volts at 1.0 kHz to a 5,000 ohm to 5.0 ohm pad connected to pins J1-L and J1-M and set selector switch to PVT position and MIC switch to position 1 (5 ohms). Jumper pin J1-A to ground and measure the output. With J1-A jumpered to ground, place selector switch in all other positions except PVT to verify that the output at pin J2-g and J2-P is off. These measurements shall be made to determine compliance with 3.13.2.2.

4.10.2.3 ICS off. Apply a 1.0-kHz input signal to pins J2-R and J2-S and adjust the input level to provide a 1.25 volt signal output at pins J2-D and J2-E. Switch the HOT MIC-NORM - ICS OFF switch to the ICS OFF position. Measure the voltage across pins J2-D and J2-E to determine compliance with 3.13.2.3.

4.10.3 Sidetone. Apply 0.6 volts at 1.0 kHz to a 5000-ohm to 5-ohm pad connected to pins J1-L and J1-M and set selector switch to ICS position and MIC switch to position 1 (5 ohms). Jumper pin J1-d and then pin J1-b to ground and measure the output voltage across pins J2-D and J2-E. Jumper pin J1-A to ground, selector switch to PVT position and measure the output voltage at pins J2-D and J2-E. Remove the jumper and apply 0.39-volts reference signal to pins J2-R and J2-S and measure the output voltage. These output voltages shall be compared to determine compliance with 3.13.3.

4.10.4 Response, microphone amp. Set MIC switch to position 1 (5 ohms) and selector switch to ICS position, jumper pin J1-d to ground, apply 0.1 volt at 1.0 kHz to a 5,000 ohm to 5 ohm pad connected to pins J1-L and J1-M and using the output level obtained across pins J2-R and J2-S as a reference level, measure the frequency response from 350 to 4,500 Hz and at 87 Hz and 18,000 Hz to determine compliance with 3.13.4. Repeat above after setting MIC switch to position 2 (150 ohms) and using a 2.4K ohm to 150 ohm pad with blocking capacitor, 47 ufd or greater instead of a 5,000 ohm to 5 ohm pad.

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4.10.5 Distortion microphone amp.

4.10.5.1 Distortion microphone amp hi voltage. Apply 4.0 volts at 350 Hz and 4.5 kHz to pins J1-L and J1-M through a 5000-ohm to 5-ohm pad and set selector switch to ICS position, place all monitor switches to OFF, master volume control to maximum audio output position and MIC switch to position 1 (5 ohms) and HOT MIC switch to HOT MIC position and measure the microphone amplifier output distortion to determine compliance with 3.13.5.

4.10.5.2 Distortion microphone amp, additional. Apply 0.6 and then 4.0 volts at 1.0 kHz to pins J1-L and J1-M through a 5000 ohm to 5 ohm pad and set selector switch to ICS position, place all monitor switches to OFF, master volume control maximum CW and MIC switch to position 1 (5 ohms) and HOT MIC switch to HOT MIC position and measure the microphone amplifier output distortion at each of the input voltages at 350 Hz, 1000 Hz and 4500 Hz to determine compliance with 3.13.5. Reduce the input voltages to .085 volts at 1.0 kHz and measure the zero crossing distortion to determine compliance with 3.13.5.

4.10.6 Attack and release time, AGC. Set selector switch to ICS position and MIC switch to position 1 (5 ohms) and HOT MIC switch to HOT MIC position and apply 1.34 volts at 1.0 kHz to a 5000-ohm to 5-ohm pad connected to pins J1-L and J1-M and measure the output voltage across pins J2-R and J2-S for a reference level. Remove the input signal for 30 seconds and reapply the input signal voltage, measure the AGC attack time required for the output voltage to reach within 1.0 dB of the reference level. Lower the input signal voltage to .85 $\pm$ 10% v then reduce voltage by 16 $\pm$ 1 db and hold for 20 seconds. Measure output voltage for a reference level. Increase the input voltage by removing the 16 db attenuation. Hold for 5 seconds and then insert the attenuation. Measure the release time required for the input voltage to reach within 1 db of the reference voltage. These measurements shall be made to determine compliance with 3.13.6.

4.10.7 Breakpoint and slope, AGC. Set MIC switch to position 1 (5 ohms) and selector switch to ICS position and apply 0.1 volt at 1.0 kHz to a 5000-ohm to 5-ohm pad connected to pins J1-L and J1-M and measure the output voltage across pins J2-R and J2-S. Increase the input voltage to 0.2 volt thence to 3.0 volts and measure the output voltages. Compare the input-output voltages to determine the breakpoint. Utilizing the output voltage obtained with 0.5-volts input as a reference level, increase the input signal level to 5.0 volts and measure the output voltage, compare the input-output voltages to determine the slope. Switch the microphone selector switch to the 150 ohm position and repeat above with 2.4K to 150 ohm pad with blocking capacitor, 47 ufd or greater connected to pins J1-L and J1-M. The comparisons shall be made to determine compliance with 3.13.7.

4.10.8 Transmit audio output, microphone amp. Apply 0.6 volts at 1.0 kHz to a 5,000 ohm to 5 ohm pad connected to pins J1-L and J1-M and set selector switch to position 1, also set MIC switch to position 1 (5 ohms). Connect Pin J1-d to ground. Set the output level, adjust control to obtain 0.39 volts across pins J1-N and J1-P. Measure the output voltage at pin pairs J1-S and J1-T, J1-V and J1-W, J1-k and J1-s, and J1-p and J1-q, with selector switch set to positions 2, 3, 4, and 5. Set microphone switch to position 2, and connect a 2.4K to 150 ohm pad with blocking capacitor, 47 ufd or greater to pins J1-L and J1-M. Apply 0.6 volts at 1.0 kHz to the pad, set selector switch to position 1, and jumper pin J1-d to ground. Observe that the output across pins J1-N and J1-P is 0.39 volts $\pm$ 1dB. Measure the output

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voltage at pin pairs J1-S and J1-T, J1-V and J1-W, J1-k and J1-s, and J1-p and J1-q, with selector switch set to positions 2, 3, 4, and 5 to determine compliance with 3.13.8.

4.10.9 Audio, headset amp.

4.10.9.1 Audio output, single tone, headset amp. Set MIC switch to position 1 (5 ohms), set channel select switch to ICS. Adjust all volume controls to maximum clockwise and push all monitor switches in (On). Connect a 1.0 kHz input at 2.75 volts to pin pair J2-A,B. The audio output shall be 1.25 volts ± 5% at pin pair J2-D,E across 8 ohms. Apply a 3.9 volt and then a 5.4 volt input signal at 1.0 kHz to determine compliance with limiting threshold and maximum limited output requirements of 3.13.9. Reset input level until output is 1.25 volts RMS and then set MIC switch to position 2 (150 ohms). The audio output shall be 1.25 volts ± 10%. Return MIC switch to position 1 (5 ohms) and set input level to 2.75 volts. Then apply input in turn to following pin pairs:

J2-V,W	J2-X,Y	J2-F,G	J2-n,p	J2-J,K	J2-d,e	J2-q,r
J2-U,j	J2-k,m	J2-a,b	J2-c,H	J2-s,t	J2-L,M	J2-N,f

The audio output shall be 1.25 volts ± 5% for each of above inputs. Apply 0.39 volts RMS to pin pair J2-R,S to obtain same output and then set channel select switch to PVT and apply 0.39 volts RMS to pin pair J2-g,p to obtain same 1.25 volts ± 5% output.

4.10.9.2 Audio output, 2-tone test, headset amp. Apply a 5.4 volt input signal of 1.0 kHz to a selected input and a 2.75 volt 1.5 kHz signal to a selected nonrelated input. Observe that the 1.5 kHz output signal is not reduced in amplitude by greater than 9 dB by the presence of the 1.0 kHz signal to determine compliance with 3.13.9.

4.10.10 Response, headset amp. Set MIC switch to position 1 (5 ohms) and apply a 1.0 kHz input at 0.39 volts to pins J2-R and J2-S and adjust volume control for a 0.7 volt output across pins J2-D and J2-E and using this output level as a reference level while maintaining a constant input level, measure the frequency response from 350 to 4,500 Hz and 87 Hz and 18,000 Hz to determine compliance with 3.13.10.

4.10.11 Distortion, headset amp.

4.10.11.1 Distortion, 350 Hz, headset amp. With MIC switch in position 1, volume control fully CW and all monitor controls in (ON) and fully clockwise apply 0.39 volts to ICS and PVT audio inputs and 2.75 volts to each radio audio input in turn and measure the headset amp output distortion at 350 Hz for compliance with 3.13.11.

4.10.11.2 Distortion, full range, headset amp. Apply 0.39 volts to ICS and PVT audio inputs and 2.75 volts to each radio channel audio input in turn to obtain 1.25 volts ± 5% output at 1.0 kHz across pins J2-D and J2-E. Measure the headset amp output distortion at 350 Hz, 1000 Hz and 4500 Hz for compliance with 3.13.11. Reduce the input voltage to .055 volts for ICS and PVT audio inputs and to 0.39 volts for each radio channel audio input and measure the zero crossing distortion to determine compliance with 3.13.11.

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4.10.12 Broadband noise, headset. Set the selector switch to ICS position and all received audio monitor switches to the ON position. With no received audio, ICS or PVT input signals, measure the broadband noise across pins J2-D and J2-E to determine compliance with 3.13.12.

4.10.13 Isolation, monitor switches/audio lines. Set selector switch to position 1, all monitor switches out (OFF) and maximum CW, HOT MIC switch to NORM and MASTER volume to maximum CW. Apply a 1.0 kHz signal at 2.75 volts to pin pair J2-A,B to obtain 1.25V ± 5% output across J2-D,E. Set selector switch to ICS position and measure the output level relative to the 1.25 volt output to determine compliance with 3.13.13. In turn, apply the input to pin pairs J2-V,W and J2-U,j and J2-X,Y and J2-k,m and J2-F,G and J2-a,b and J2-n,p and J2-c,H and J2-J,K and J2-s,t. As each input is applied, measure the output level relative to the 1.25 volt output to determine compliance with 3.13.13. Set selector switch to position 2, apply above input to J2-V,W to obtain 1.25V ± 5% output. Set selector switch to position 3, apply above input to J2-U,j to obtain 1.25V ± 5% output. Set selector switch to position 4, apply above input to J2-X,Y to obtain 1.25V ± 5% output. Set selector switch to position 5, apply above input to J2-k,m to obtain 1.25V ± 5% output. Leave selector switch in position 5 and apply 0.39 volt signal input to pin pair J2-R,S to obtain 1.25V ± 5% output. Set Hot MIC switch to ICS OFF and measure the output relative to 1.25V to determine compliance with 3.13.13. Return Hot MIC switch to NORM and push all monitor switches in (ON) and apply 2.75V, 1.0 kHz to a radio input pair and measure the output level at the other radio input pin pairs. Repeat these measurements until all audio lines have been tested to determine compliance with 3.13.13.

4.10.14 Isolation, transmitter lines. Apply 0.39 volts at 1.0 kHz to pin pair J2-R and J2-S and obtain 1.25 volts at J2-D, E. Set MIC switch to position 1 (5 ohms) and selector switch to positions 1 through 5 and measure crosstalk, with and without pin J1-d jumpered to ground, at pin pairs J1-N and J1-P, J1-S and J1-T, J1-V and J1-W, J1-K and J1-s, and J1-p and J1-q, respectively, while applying in turn 0.39 volts to ICS and PVT inputs and 2.75 volts to all monitor inputs except that corresponding to the selected channel. Also measure at pins J1-f, J1-g, J1-h, J1-r and J1-t, terminated in 270 ohms to chassis ground, as selector switch is set to positions 1 through 5, respectively, and pin J1-d is jumpered to ground for each input.

Apply 0.6 volts at 1.0 kHz to a 5,000 ohm to 5 ohm pad connected to pins J1-L and J1-M. Set MIC switch to position 1 and selector switch to position ICS, PVT and 1 through 5 and measure crosstalk on all transmit audio pairs, except that corresponding to the selected channel, with the appropriate key line jumpered to ground. Now set MIC switch to position 2 and measure crosstalk on all unselected transmit audio pairs at each position of the selector switch while applying 0.6 volts at 1.0 kHz to a 2,400 ohm to 150 ohms pad connected to pins J1-L and J1-M with the appropriate key line jumpered to ground. The crosstalk levels at the transmitter audio and transmitter control lines shall comply with 3.13.14.

4.10.15 Transient protection. Under standard test conditions, apply a +80-volt pulse and -80-volt pulse for 0.2-second duration between pins J1-C and J1-D. Apply each pulse three (3) times within one (1) minute as specified in MIL-STD-704, Category "B." Perform both of these tests to determine compliance with 3.13.15. In addition, apply at least three (3) aperiodic spike waveforms between Pins J1-C and

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J1-D. The spike waveform, produced by a generator with a source impedance of 50 ± 50 ohms, shall satisfy the following requirements:

Open-circuit voltage: ±600-volts peak
Risetime: 0.9 ± 0.2 microseconds
Falltime: 10.0 ± 1.0 microseconds

4.10.16 Headset amp, unterminated. With input power applied and input signals removed, disconnect the 8-ohm load across pins J2-D and J2-E. Monitor the amplifier output each time the switches and volume controls are operated. Short pins J2-D and J2-E to ground for 1 minute. Observe that no unusual emanations occur from the unit while shorted and that no damage is sustained when the short is removed, to determine compliance with 3.13.16.

4.10.17 Transmitter control.

4.10.17.1 Transmitter control, 1 amp. Jumper pins J1-c and J1-d, set selector switch to position 1, and apply a plus dc voltage through a current limiting device to pin J1-f (negative to pin J1-D) to obtain a 1.0-ampere current flow. In turn, set selector switch to positions 2 through 5 as the dc voltage is applied to pins J1-g, J1-h, J1-r and J1-t. These current measurements shall be made to determine compliance with 3.13.17.

4.10.17.2 Transmitter control, GO/NO GO. Follow procedure of 4.10.17.1 except a GO/NO GO test may be made using a low current source.

4.10.18 Function selector logic.

4.10.18.1 Function selector logic, 1 amp. Jumper pins J1-c and J1-A, set selector switch to PVT, and apply a plus dc voltage through a current limiting device to pin J1-z (negative to pin J1-D) to obtain a 1.0-ampere current flow. Jumper pins J1-c and J1-d. In turn, set selector switch to positions ICS, and 1 through 5 as the dc voltage is applied to pins J1-Y, J1-Z, J1-i, J1-m and J1-n. These current measurements shall be made to determine compliance with 3.13.18.

4.10.18.2 Function selector logic, GO/NO GO. Follow procedure of 4.10.18.1 except a GO/NO GO test may be made using a low current source.

4.10.19 Microphone power. Set microphone selection switch in position 2, attach a 1000 ohm resistive load to pins J1-L and J1-M. Measure the dc voltage across the resistive load to determine compliance with 3.13.19.

4.10.20 Received audio input attenuation during transmit. Set selector switch to PVT position, all monitor switches to ON position, and HOT MIKE switch to NORM. Apply a 0.39-volt signal input at 1.0 kHz to pin pair J2-g and J2-P. Set MIC switch to position 1 (5 ohm) and obtain 1.25 volts ± 5% at pin pair J2-D,E. Apply a ground to pin J1-A and measure the attenuation of the headset amplifier output at pins J2-D and J2-E with respect to the normal 1.25 volt output. In turn, apply the proper input, (0.39 volts to J2-R and J2-S, J2-g and J2-P) (2.75 volts to all other audio inputs), to all audio input pairs. Measure the headset amplifier output attenuation, apply a ground to pin J1-d and repeat this test for all conditions described in Table II to determine compliance with 3.13.20.

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4.10.21 Audio level (volume adjustments).

4.10.21.1 Main volume control. Connect a 1.0-kHz input signal successively to each controlled received audo input (i.e., PVT, ICS, 1, 2, 3, 4, 5, NAV 1, NAV 2, DIR) and adjust the level to obtain 1.25 volt output across J2-D and J2-E, with both the individual monitor (where applicable) and master audio level controls set for maximum audio output. Adjust the main volume control for minimum audio output and measure this voltage for compliance with 3.13.21.1.

4.10.21.2 Monitor audio level adjustment. Connect a 1.0-kHz input signal successively to each switched received audio input (i.e., 1, 2, 3, 4, 5, NAV 1, NAV 2), and adjust the level to obtain 1.25 volt output across J2-D and J2-E, with both the individual monitor and master audio level controls set for maximum audio output. Adjust the appropriate individual audio level control for minimum audio output and measure this voltage for compliance with 3.13.21.2.

4.10.22 TEMPEST testing. The equipment shall be tested in accordance with the procedures of Appendix A and the contractor prepared, Government approved TEMPEST Test Plan.

4.11 Electromagnetic interference. The equipment shall be tested for compliance with the requirement of 3.14. Test setups and measurements techniques shall be in accordance with MIL-STD-462, Notice 2, as detailed in a contractor prepared, Government approved test plan. The unit(s) controls shall be adjusted as defined except as may be required to simulate special conditions. Selector switch to ICS, Hot MIC switch to Hot MIC, all monitor switches in (ON), and MIC switch to position 1 (5 ohms), volume controls full clockwise position. The length of interconnecting cable shall be approximately six (6) feet. All emission tests shall be conducted with maximum input/output signal processed through the audio channels.

4.11.1 Audio susceptibility, CS 01 (AVC hold). The unit(s) shall be tested in accordance with Method CS 01. The interphone output J2-R/J2-S shall be monitored with a scope and any deviation from normal output ± 3 dB shall be recorded. In addition to the above, the noise output on headset and interphone lines shall be measured. These tests and measurements shall be made to determine compliance with 3.14.1.

4.11.2 Conducted susceptibility, CS 02. The unit(s) shall be tested in accordance with Method CS 02. The requirements of 3.14.2 shall be met.

4.11.3 Radiated susceptibility, RS 03. The unit(s) shall be tested in accordance with Method RS 03. The requirements of 3.14.3 shall be met.

4.11.4 Radiated susceptibility (induction field). Set the unit(s) to be tested six (6) inches directly above a wire which shall be placed parallel to the front of the unit and 1.25 inches behind the front panel. Connect the wire to a 400-Hz power source and adjust the current to 5 amps, rms. Set the unit controls as in 4.10.2 and 4.10.3 and measure the output levels to determine compliance with 3.14.4.

4.12 Interchangeability. The mechanical dimensions and electrical performance of parts shall be physically and electrically interchangeable among the unit(s). Mechanical dimensions and electrical performance required by the guage drawings and

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equipment drawings shall be measured to determine conformance with physical and functional interchangeability requirements of 3.19. When a listed dimension or performance requirement is not within specified limits, it shall be considered a major defect. The following dimensions or performance shall be gauged or measured.

a. Mechanical (dimensions). The overall dimensions and mounting dimensions shall apply.

b. Electrical performance (assemblies). Shall be tested to determine compliance with the applicable performance requirements specified on the equipment drawings.

4.13 Operational inspections

4.13.1 Operational inspections (with Test Set). Each unit(s) shall be connected to Test Set TS-3947( )/AR or demonstrated equivalent to form an operational system. A talk test shall be performed to determine compliance with 3.19. The unit(s) shall be operated in all modes of operation. All switches and volume controls shall be operated and perform their normal functions.

4.13.2 Operational Inspection (system simulation). Each unit(s) shall be connected with at least nine other units (C-10414(V)( )/ARC) to form an operational system. (Substitution of simulated additional units is permissable when operational conditions are accurately duplicated. The substituted fixture shall be approved by the procuring activity.) A talk test shall be performed to determine compliance with 3.19. The unit(s) shall be operated in all modes of operation. All switches and volume controls shall be operated and perform their normal functions.

4.14 Mechanical and visual inspections. The unit(s) shall be examined to determine conformance to the requirements of this specification including markings; accuracy of dimensions; quality of workmanship; maximum weight allowed, use of proper materials, parts, and processes; freedom from visible defects or any other imperfections that would result in rejection of the unit; and requirements 1, 5, 7, 8 and 9 of MIL-STD-454. All moving parts shall be examined to ensure that they operate freely without sticking or binding, and yet fit with the accuracy required by this specification and the applicable drawing. Defects shall be classified in accordance with MIL-STD-252.

4.14.1 Weight. The unit(s) shall be weighed to determine compliance with 3.2.1.

4.15 Quality conformance inspection of packaging. Packaging shall be tested in accordance with MIL-P-116 to determine conformance to the requirements of Section 5. The rough handling test will be performed only when specified (see 6.2c).

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B or Commercial, as specified in contract (see 6.2b).

5.1.1 Level A.

5.1.1.1 Cleaning. Control Unit, Communication System C-10414(V)( )/ARC shall be cleaned in accordance with process C-1 of MIL-P-116.

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5.1.1.2 Drying. Control Unit, Communication System C-10414(V)( )/ARC shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservative application. None required.

5.1.1.4 Unit packing. Unit packing shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature. Each technical literature shall be unit packed method IC-1 of MIL-P-116.

5.1.1.4.2 Packaging. Control Unit, Communication System C-10414(V)( )/ARC shall be individually unit packed Method III of MIL-P-116 as follows: Cushion the unit on all surfaces with cells or pads, or both, fabricated of fiberboard conforming to PPP-F-320, Type CF, class weather-resistant, variety SW, Grade W5c, designed to protect all projections and absorb the shock of impact in handling and transit. As an alternate, cushioning material may be fabricated of plastic molding material conforming to MIL-P-19644 or polyurethane form conforming to MIL-P-26514. Place each cushioned unit within a close-fitting fiberboard box conforming to PPP-B-636, W5C. Place the technical literature, unit packed as specified in 5.1.1.4.1, on top of the contents directly below the lid of the box. Close the box in accordance with the appendix of the box specification.

5.1.2 Level B. Cleaning, drying, preservative application and unit packing shall be as specified in 5.1.1.

5.1.3 Commercial preservation. Preservation shall be in accordance with ASTM D 3951.

5.2 Packing. Packing shall be Level A, B or Commercial, as specified (see 6.2b). Shipping containers for Level A and B shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents.

5.2.1 Level A. A quantity of Control Unit, Communication System C-10414(V)( )/ARC, unit packed as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, Class 2, or PPP-B-585, style 2 or 3, Class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4-inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, Type I, Finish A.

5.2.1.1 Unitization. Palletization shall be required when: containers specified in 5.2.1 do not require skids; quantities per destination comprise a pallet load; and container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of containers, packed as specified in 5.2.1, except that container strapping may be omitted, shall be placed on a pallet, load Type I, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, Type IV, Group I or II woods.

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The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, Type I, Finish A, or shrink film.

5.2.2 Level B. A quantity of Control Unit, Communication System C-10414(V)( )/ ARC, unit packed as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-640, class 2, style E, or PPP-B-636, type CF, class weatherresistant. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-0097 and PPP-S-760, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, containers will be pallet-mounted on pallets conforming to NN-P-71, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film. When a single unit is shipped to a single destination no packing is required; the unit container shall serve as the shipping container. Reinforcing shall be as specified above.

5.2.2.1 Unitization. Palletization shall be required when quantities per destination comprise a pallet load. A quanity of Control Unit, Communication System C-10414(V)( )/ARC, unit packed as specified in 5.1, shall be placed on a pallet, load type I, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, of shrink film.

5.2.3 Commercial packing. Packing shall be in accordance with ASTM D 3957.

5.3 Marking.

5.3.1 Military marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping cortainers shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with ASTM D 3957.

6. NOTES.

6.1 Intended use. Control Unit, Communication System C-10414(V)( )/ARC is intended for use by the pilot, copilot or crew to provide control of radio equipments for voice transmission and reception, direction finding equipment and intercommunication when installed in aircraft.

6.2 Ordering data. Procurement documents shall specify the following:

a. Title, number and date of this specification and any amendment thereto.

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b. Level A, level B or Commercial preservation and packing (see section 5).

c. When rough handling is required.

d. First article inspection:

(1) Several first article unit(s) are generally required so that lengthy environmental tests can be completed on specified sample units while complete performance measurements can be made on one sample unit (see 3.3).

(2) First article pack(s) as follows:

Makeup of pack(s).

Number of each kind of pack to be submitted.

Inspection to be performed thereon (including rough handling).

e. Marking and shipment of sample of first article samples and production unit(s).

f. Place of final inspection.

g. Submission of the statement of treatment referenced in 3.10.

6.3 Group C inspection. Approval to ship unit(s) on order may be withheld, at the discretion of the Government pending decision from the procuring activity, until compliance with 4.5.4.1 is complete.

6.4 Nomenclature. The nomenclature assigned to the equipment covered by this specification is as follows:

a. Control Unit, Communication System C-10414(V)1/ARC, consisting of Control Unit, Communication System C-10414(P)/ARC assembled with Panel, Illuminated, RED, 28V, MX10516/ARC.

b. Control Unit, Communication System C-10414(V)2/ARC, consisting of Control Unit, Communication System C-10414(P)/ARC assembled with Panel, Illuminated, RED, 5V, MX10515/ARC.

c. Control Unit, Communication System C-10414(V)3/ARC, consisting of Control Unit, Communication System C-10414(P)/ARC assembled with Panel, Illuminated, GRN, 115V, MX10517/ARC.

6.5 Deviations. Request for deviations in packaging and packing design shall be supported by test data proving the ability of the equipment, packaged and packed in accordance with the proposed design change, to withstand the applicable tests for the selected method, including the rough handling tests in accordance with MIL-P-116. Deviation approval shall be contingent on availability of packaging and packing materials selected and economics realized (see Section 5).

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6.6 Environmental. Environmental pollution measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.7 Definitions.

6.7.1 Branching. Branching is a connected arrangement of filaments (hyphae) formed by shoots or secondary stems growing from the main stem or filament (hyphae).

6.7.2 Growth colonization. Growth colonization is a mass of individual plants, generally of one species, living together; or a group of hyphae which is formed from one spore or cell and may be one individual plant. Colonization which completely covers the surface of the nutrient material constitutes abundant growth.

6.7.3 Microbial growth. Microbial growth is the growth of very minute organisms. Such organisms when present in large numbers may provide a colony visible t the naked eye.

6.7.4 Sporulation. Sporulation is the formation of minute unicellular reproductive or dormant bodies, called spores.

6.7.5 Tubular germination. Tubular germination is partial growth by the production of hyphae, which are tubular shaped fungal filaments. Tubular germination constitutes restricted individual spore growth not proceeding to colonization.

6.8 Subject term (key word) listing.

Intercommunication

6.9 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Army (AV)
(Project 5821-A152)

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APPENDIX A

TEMPEST REQUIREMENTS

FOR THE

CONTROL UNIT, COMMUNICATION SYSTEM C-10414(V)( )/ARC

NOTE: Appendix A, entitled as above, is classified CONFIDENTIAL and is enclosed under separate cover, as required.

APPLICABLE DOCUMENTS

The following documents, of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein.

MIL-STD-831	Preparation of Test Reports
NACSEM 5100	Compromising Emanations Laboratory Test Standard, Electromagnetics (Document CONFIDENTIAL, Title Unclassified)

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both title and number or symbol should be stipulated when requesting copies.)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-49227A (AV)	Control Unit, Communication System C-10414 (V) ( )/ARC

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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